IoT Architecture Reference Layer

- A reference architecture serves as a standardized blueprint that provides a clear structure and guidelines for designing and implementing an IoT system.
- By using a reference architecture, developers can reduce design complexity, ensure interoperability, and accelerate the development process, ultimately leading to more efficient and reliable IoT solutions.
- IoT reference architectures typically consist of multiple layers that work together to enable the functioning of an IoT system.

Cont…

- **Perception Layer:**This layer comprises the physical devices or sensors that collect data from the environment or interact with the physical world. These devices can include temperature sensors, motion detectors, cameras, and other IoT−enabled devices.
- **Network Layer:** The network layer facilitates the connectivity and communication between the IoT devices and the cloud or other data processing components. It includes protocols, gateways, routers, and other networking infrastructure to ensure seamless data transfer and reliable connections.
- **Data Processing Layer:** This layer involves processing and analyzing the data collected from IoT devices. It may include edge computing devices or cloud−based platforms where data is aggregated, filtered, transformed, and analyzed to derive valuable insights.
- **Application Layer:** The application layer encompasses the software applications or services that utilize the processed IoT data to provide specific functionalities or address specific use cases. These applications can range from real−time monitoring and control systems to predictive analytics, machine learning algorithms, and automation

Key Components of IoT Reference Architecture

- **Devices:** These physical objects, ranging from sensors to smart appliances, play a crucial role in collecting data and transmitting it to the cloud.
- **Network:** The network infrastructure, which can be wired or wireless, connects the devices to the cloud, enabling seamless data transfer and communication.
- **Cloud:** The cloud serves as the centralized storage and processing hub for the data collected by the devices. It can be a public, private, or hybrid cloud, depending on specific requirements.
- **Applications:** Software applications leverage the data collected by devices to deliver insights and value. These applications can be tailored for various purposes, including monitoring, control, and automation.
- **Security:** Security measures are essential to protect the data collected by devices from unauthorized access. Techniques like encryption, authentication, and authorization ensure data security.
- **Privacy:** Privacy measures safeguard personal data collected by devices from unauthorized access.
- **Governance:** Governance policies ensure responsible and ethical use of the IoT system. These policies address aspects such as data ownership, data sharing, and appropriate data usage.

Benefits of Using IoT Reference Architecture

- **Common Framework:** IoT reference architecture provides a standardized framework for designing and implementing IoT solutions, ensuring consistency and interoperability across systems.
- **Security and Scalability:** The architecture serves as a foundation for implementing robust security and scalability measures, safeguarding IoT systems against threats and enabling future growth.
- **Cost Efficiency:** By using a reference architecture, organizations can avoid reinventing the wheel and utilize existing technologies and expertise, reducing the cost of development and deployment.
- **Faster Time to Market:** Utilizing a reference architecture accelerates the implementation of IoT solutions, enabling organizations to get their systems up and running more quickly and efficiently.

Examples of IoT Infrastructure

- Connected cars.
- Smart appliances.
- Connected security systems.
- Smart agriculture equipment.
- Connected retail.
- Connected healthcare monitors.
- Connected manufacturing equipment.
- Connected cities.

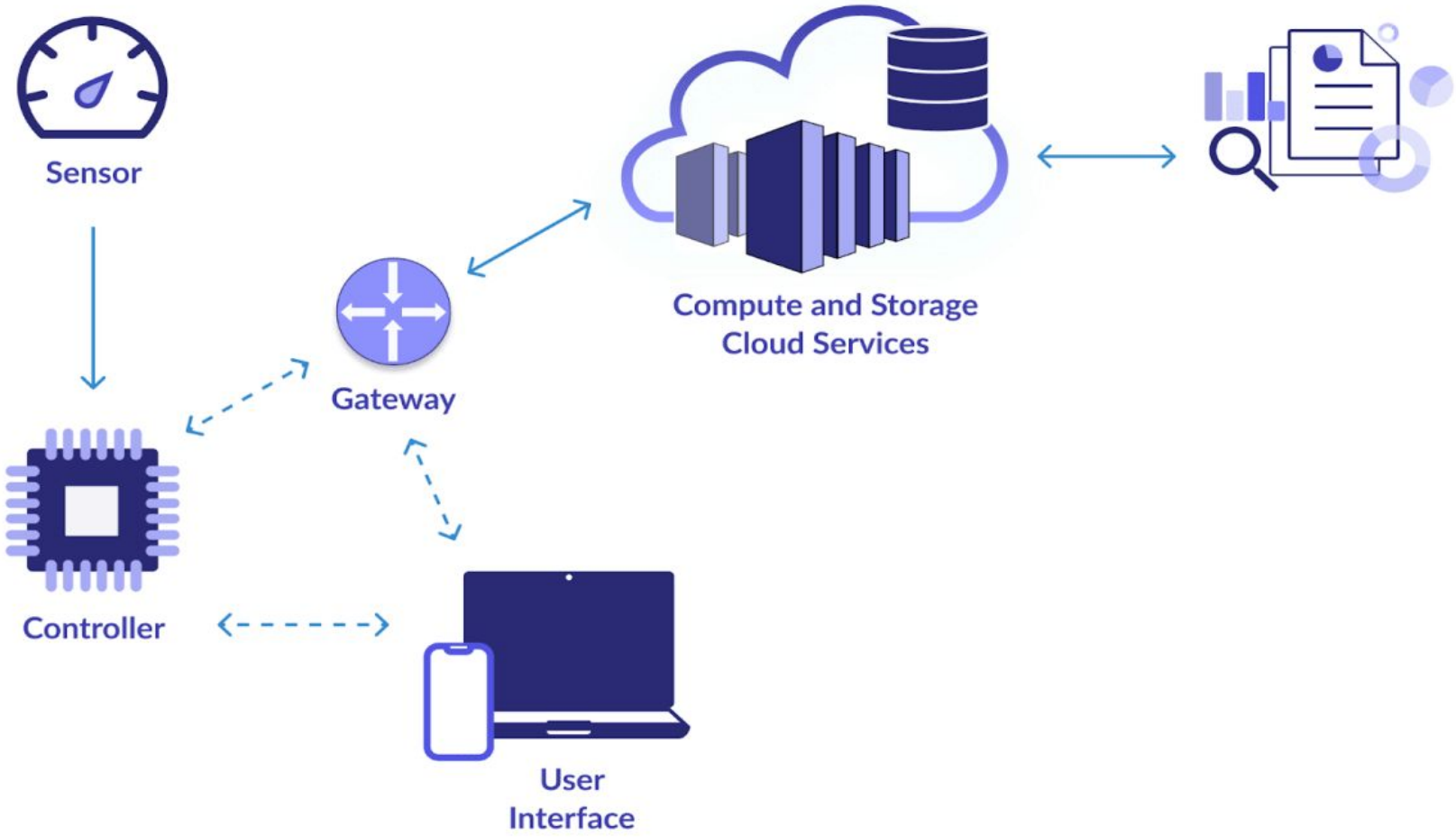
Source: https://builtin.com/articles/iot-examples

Elements of IoT infrastructure

- IoT refers to any system of interconnected devices that have sensors and embedded processing abilities.
- **Note that devices don't have to use the Internet.**
- Even locally connected devices interacting and exchanging data are an IoT system.
- With that in mind, we can break down the IoT infrastructure required to make an IoT system into the following elements.

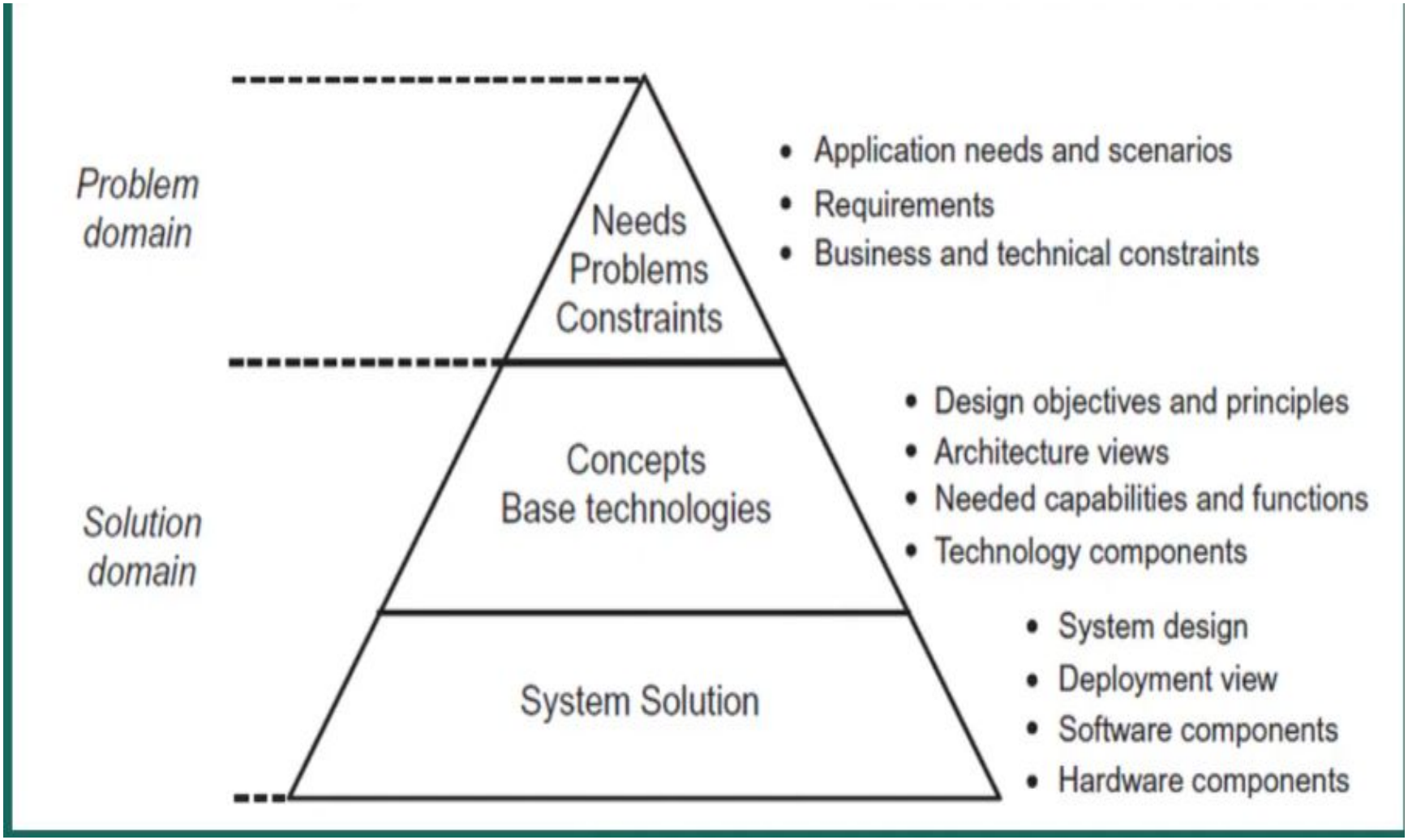
IoT Infrastructure Element	Description
Sensor	Used for measuring physical quantities that the IoT device shares over the network
Controller	The brain of the device; acts as a bridge between the sensor and the network and also performs onboard computations and storage
Network	The technology used for exchanging data with either other devices in the system, or the Cloud
Cloud	Computing, storage and gateway resources accessible over the internet
User-facing applications	Mobile and web applications that allow the user to interact with the IoT system and visualize the data
Data Analytics	The tools and resources (often on the Cloud) that enable users to derive insights from the data sent by the IoT system

Elements of Infrastructure for an IoT System



M2M to IoT architecture

- M2M Communication allows between devices of the same time and a specific application via wireless or wired communication.
- M2M is applied in remote monitoring, or to provide connectivity of remote machine type devices.
- **However, M2M doesn't allow the connection or sharing of devices over the internet.**
- The IoT is a widely used term for the set of technologies, systems and design principles associated with the emerging wave of internet-connected things that are based on the physical environment.



M2m to IoT Architecture

Bluetooth

- Bluetooth is used for short-range wireless voice and data communication.
- It is a Wireless Personal Area Network (WPAN) technology and is used for data communications over smaller distances.
- It operates within the unlicensed, business, scientific, and clinical (ISM) bands from 2.4 GHz to 2.485 GHz.
- Bluetooth ranges up to 10 meters. Depending upon the version, it presents information up to at least 1 Mbps or 3 Mbps.
- The spreading method that it uses is FHSS (Frequency-hopping spread spectrum).
- A Bluetooth network is called a piconet and a group of interconnected piconets is called a scatter net.

Architecture of Bluetooth

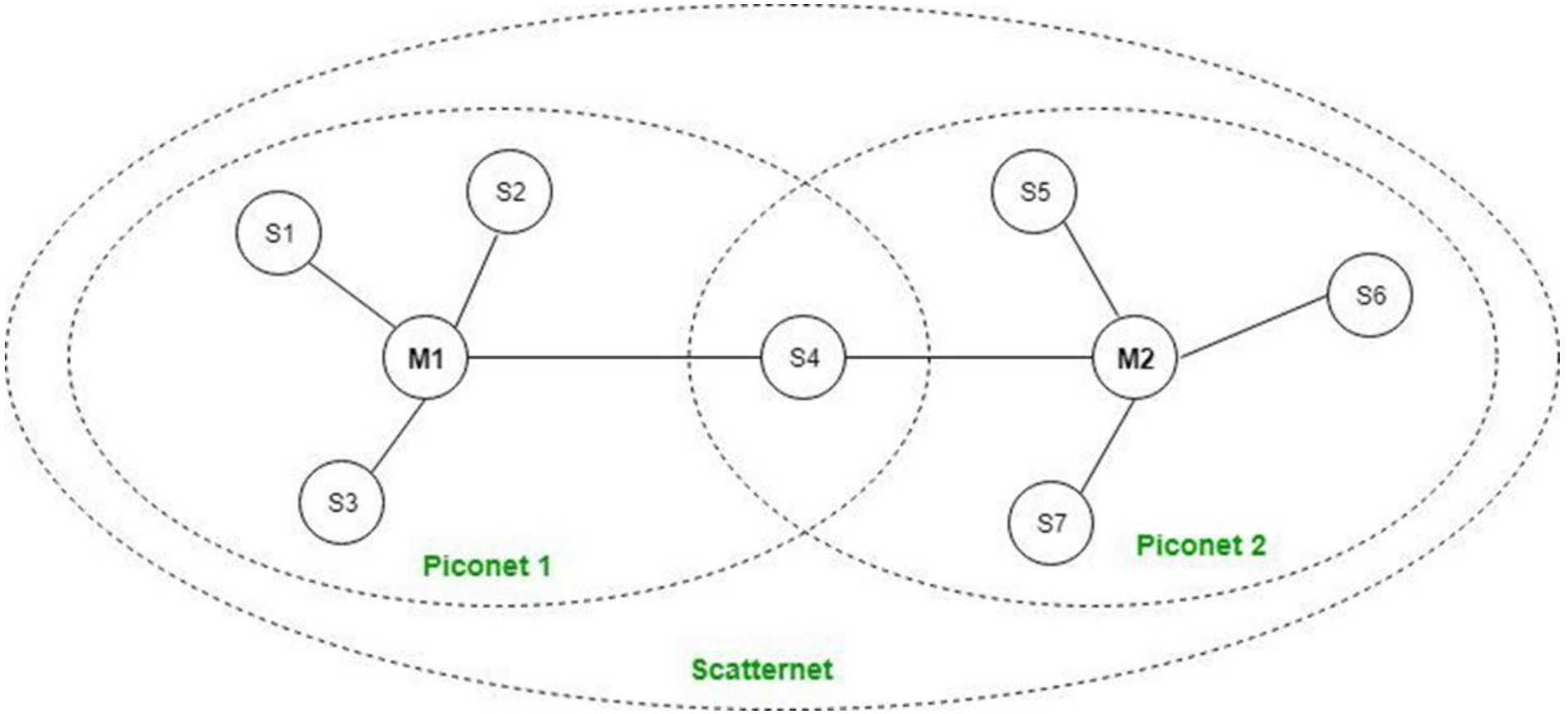
The architecture of bluetooth defines two types of networks:

1. Piconet

- Piconet is a type of Bluetooth network that contains one primary node called the master node and seven active secondary nodes called slave nodes.
- Thus, we can say that there is a total of 8 active nodes which are present at a distance of 10 meters.
- The communication between the primary and secondary nodes can be one-to-one or one-to-many.
- **Possible communication is only between the master and slave; Slave-slave communication is not possible.**

2. Scatternet:

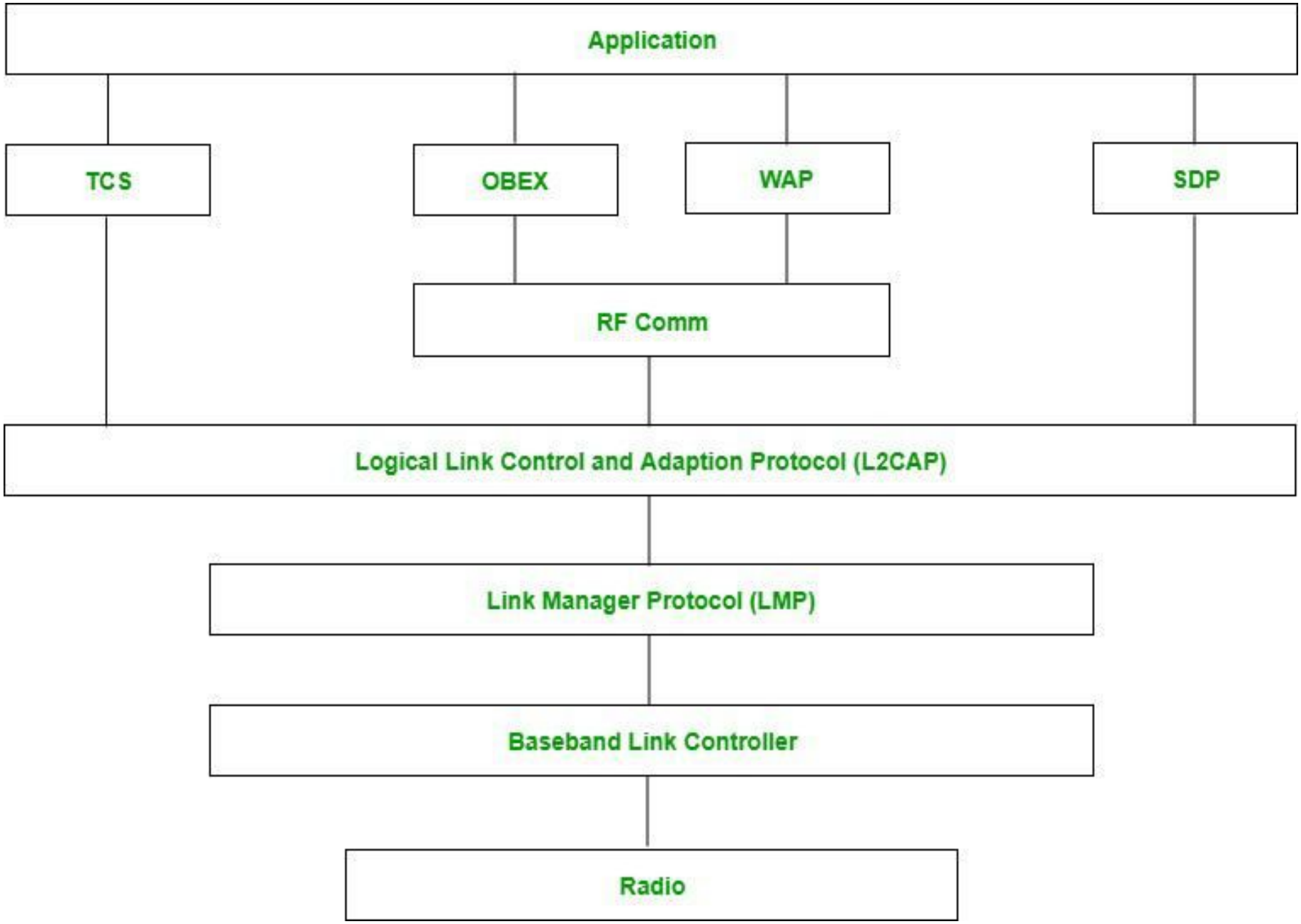
- It is formed by using various piconets.
- A slave that is present in one piconet can act as master or we can say primary in another piconet.
- This kind of node can receive a message from a master in one piconet and deliver the message to its slave in the other piconet where it is acting as a master.
- This type of node is referred to as a bridge node. A station cannot be mastered in two piconets.


S1
S2
S3
M1
S4
S5
M2
S6
S7
Piconet 1
Piconet 2
Scatternet

Bluetooth Protocol Stack

- **Radio (RF) Layer:** It specifies the details of the air interface, including frequency, the use of frequency hopping and transmit power. It performs modulation/demodulation of the data into RF signals.
- **Baseband Link Layer:** The baseband is the digital engine of a Bluetooth system and is equivalent to the MAC sublayer in LANs. It performs the connection establishment within a piconet, addressing, packet format, timing and power control.
- **Link Manager Protocol Layer:** It performs the management of the already established links which includes authentication and encryption processes. It is responsible for creating the links, monitoring their health, and terminating them gracefully upon command or failure.
- **Logical Link Control and Adaption (L2CAP) Protocol Layer:** It is also known as the heart of the Bluetooth protocol stack. It allows the communication between upper and lower layers of the Bluetooth protocol stack. It packages the data packets received from upper layers into the form expected by lower layers. It also performs segmentation and multiplexing.

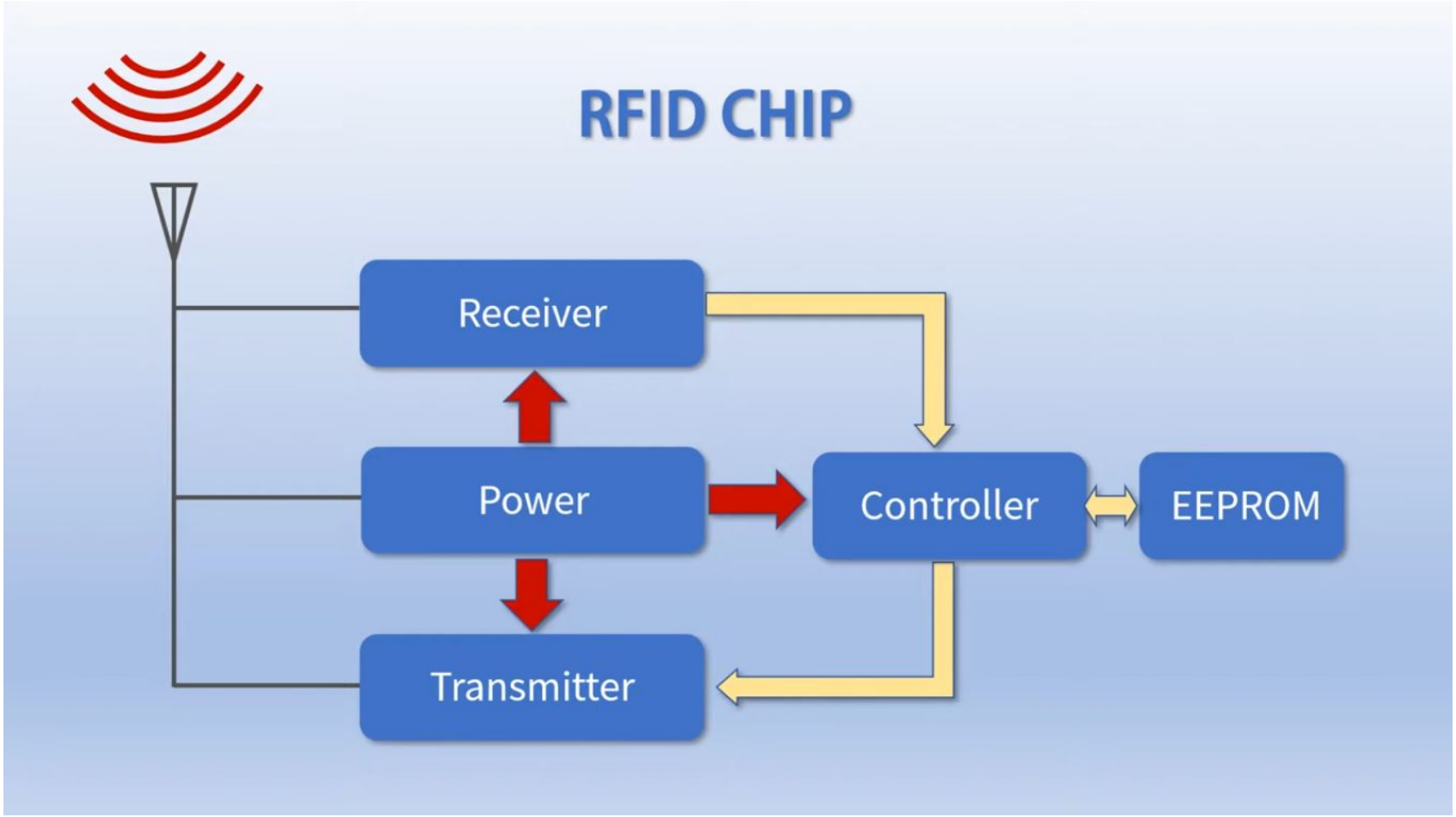
- **Service Discovery Protocol (SDP) Layer:** It allows discovering the services available on another Bluetooth-enabled device.
- **RF Comm Layer:** It is a cable replacement protocol. It provides a serial interface with WAP (wireless application protocol) and OBEX (object exchange). It also provides emulation of serial ports over the logical link control and adaption protocol(L2CAP).
- **OBEX:** It is short for Object Exchange. It is a communication protocol to exchange objects between 2 devices.
- **WAP:** It is short for Wireless Access Protocol. It is used for internet access.
- **TCS:** It is short for Telephony Control Protocol. It provides telephony service. The basic function of this layer is call control (setup & release) and group management for the gateway serving multiple devices.
- **Application Layer:** It enables the user to interact with the application.

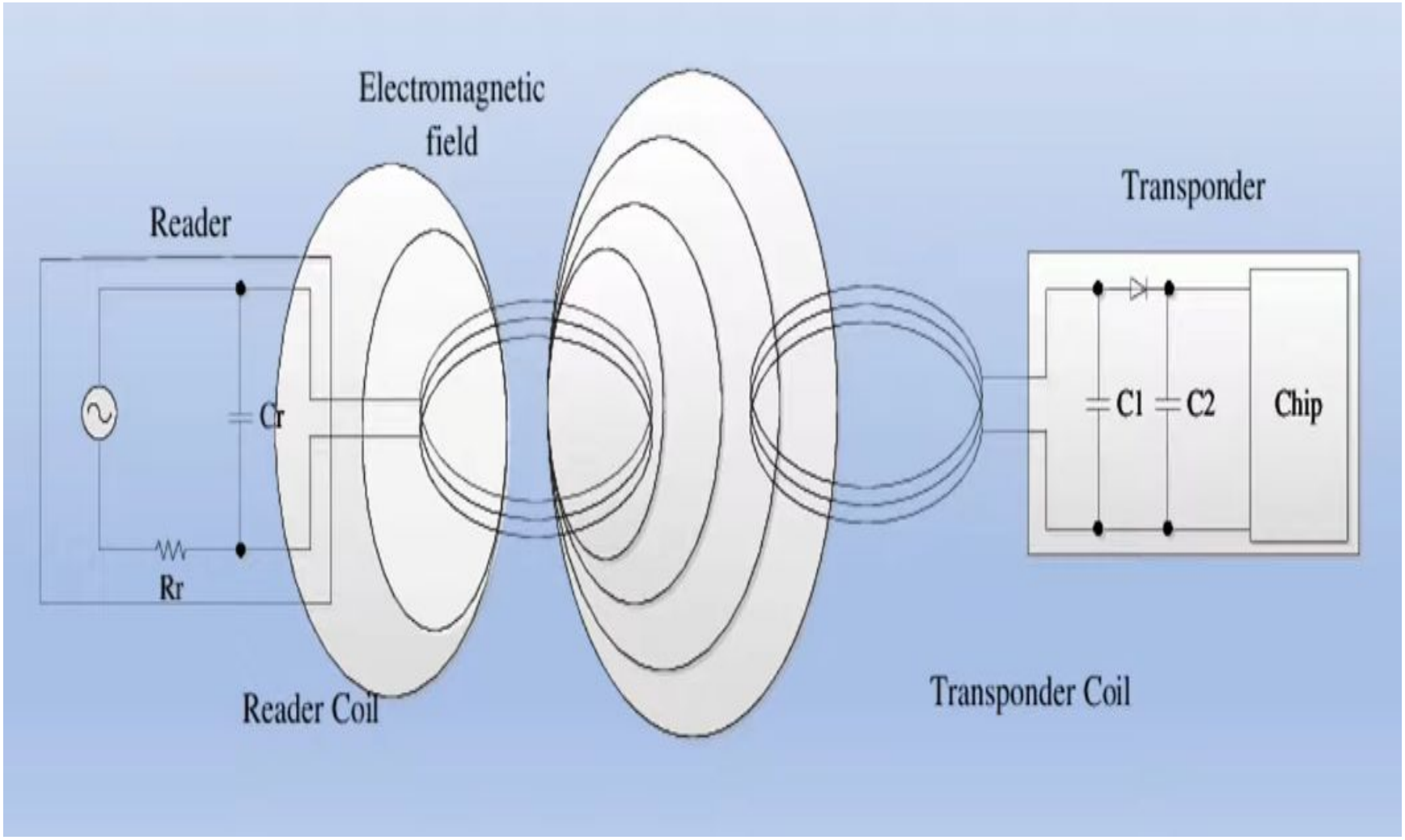

Application
TCS
OBEX
WAP
SDP
RF Comm
Logical Link Control and Adaption Protocol (L2CAP)
Link Manager Protocol (LMP)
Baseband Link Controller
Radio

- **Applications of Bluetooth**
- It can be used in wireless headsets, wireless PANs.
- It can connect a digital camera wireless to a mobile phone.
- It can transfer data in terms of videos, songs, photographs, or files from one cell phone to another cell phone or computer.
- It is used in the sectors of Medical healthcare, sports and fitness, Military.
- **Advantages**
- It is a low-cost and easy-to-use device.
- It can also penetrate through walls.
- It creates an Ad-hoc connection immediately without any wires.
- It is used for voice and data transfer.
- **Disadvantages**
- It can be hacked and hence, less secure.
- It has a slow data transfer rate of 3 Mbps.
- Bluetooth communication does not support routing.

RFID

- **Radio Frequency Identification (RFID)** is a technology that uses electromagnetic fields to automatically identify and track tags attached to objects.
- These tags contain electronically stored information that can be read from several meters away, without requiring direct line-of-sight.
- RFID is commonly used in inventory management, asset tracking, access control, and supply chain logistics due to its efficiency and accuracy in tracking and managing items.
- It is a method that is used to track or identify an object by radio transmission.
- Data is digitally encoded in an RFID tag which might be read by the reader.
- This device works as a tag or label during which data is read from tags that are stored in the database through the reader as compared to traditional barcodes and QR codes.


RFID CHIP
Receiver
Power
Controller
EEPROM
Transmitter



Circuit of a passive RFID system

Frequency of Operation

LF
(Low Frequency)
125 kHz or 134 kHz

Range: up to 10 cm

HF
(High Frequency)
13.56 MHz

Range: up to 1 m

UHF
(Ultra High Frequency)
860- 960 MHz

Range: 10 to 15 m

Types of RFID TAGS

1. Passive Tags

- No built-in power source, relying on the RFID reader.
- Less expensive, longer lifespan, shorter read range (up to a few meters).

2. Active Tags

- Have their own power source (battery), allowing for a longer read range (up to several meters).
- More expensive, limited lifespan due to the battery.

3. Semi-Passive Tags

- Small battery powers the tag's circuitry.
- Middle ground in terms of cost, range, and lifespan

- **Frequency Bands**
- RFID systems operate in different frequency bands, each with its characteristics:
- **Low Frequency (LF, 125-134 kHz)**: Used for short-range applications (up to 10 cm). Common in animal tracking and access control.
- **High Frequency (HF, 13.56 MHz)**: Medium range (up to 1 meter) and commonly used in smart cards, ticketing, and library systems.
- **Ultra-High Frequency (UHF, 860-960 MHz):** Longer range (up to 12 meters) and used in inventory management, supply chain, and logistics.
- **Microwave Frequency (2.45 GHz):** Used for very specific applications with ranges similar to UHF but with higher data transfer rates.

- **Challenges of RFID**
- **Cost**: Higher initial setup and tag costs compared to traditional barcodes.
- **Interference**: Susceptible to interference from metal, liquids, and other RFID systems.
- **Privacy**: Concerns about unauthorized tracking and data breaches.
- **Standardization**: Different frequency standards and protocols can lead to compatibility issues.
- It takes longer to program RFID Devices.
- There is privacy concern about RFID devices anybody can access information about anything.
- Active RFID can costlier due to battery.

QR CODE VS RFID: DIFFERENCES

QR CODE		RFID
Scanned Using a Smartphone Camera or a Dedicated QR Code Reader	Reading Method	Transmit Data Wirelessly to an RFID Reader Using Radio Waves
Two-Dimensional Barcode System	Technology Used	Radio Waves
Ranges From a Few Bytes to Several Kilobytes	Data Storage	Ranging From a Few Bytes to Several Kilobytes or Even More
Requires Close Proximity Between the Scanner and the Code	Range and Interaction	Can Be Read From a Few Centimetres to Several Meters Away
Relatively Simple and Inexpensive to Implement; They Can Be Printed on a Piece of Paper or Displayed on a Screen	Cost and Complexity	RFID Tags, Especially Active Ones, Are More Expensive and Complex to Implement, Requiring Specialized Hardware for Reading and Writin
Holds Less Data Than RFID Tags	Capacity and Speed	RFID Tags Can Typically Hold More Data and Be Read More Quickly and Simultaneously (A Process Called "Bulk Reading")
Used for Ticketing, Inventory Management System, Tracking Shipments, etc	Application and Usage	Used in Inventory Tracking, Supply Chain Management, Access Control, Contactless Payment Systems, Asset Tracking, Animal Tracking, etc
Scalable and Can Be Produced in Large Quantities at a Low Cost	Cost and Scalability	Cost Depends on the Type of RFID System, Tag Functionality, and Required Range
Can Be Easily Replicated or Tampered	Security and Privacy	Can Be Encrypted, and Data Can Be Protected

TRITON

802.15.4e & 802.15.4g

- **802.15.4e for industrial applications and 802.15.4g for the smart utility networks (SUN)**
- The 802.15.4e improves the old standard by introducing mechanisms such as time slotted access, multichannel communication and channel hopping.

- **IEEE 802.15.4e introduces the following general functional enhancements**:

1. Low Energy (LE): This mechanism is intended for applications that can trade latency for energy efficiency. It allows a node to operate with a very low duty cycle.

2. **Information Elements (IE)** It is an extensible mechanism to exchange information at the MAC sublayer.

3. Enhanced Beacons (EB): Enhanced Beacons are an extension of the 802.15.4 beacon frames and provide a greater flexibility. They allow to create application-specific frames.

4. Multipurpose Frame: This mechanism provides a flexible frame format that can address a number of MAC operations. It is based on IEs.

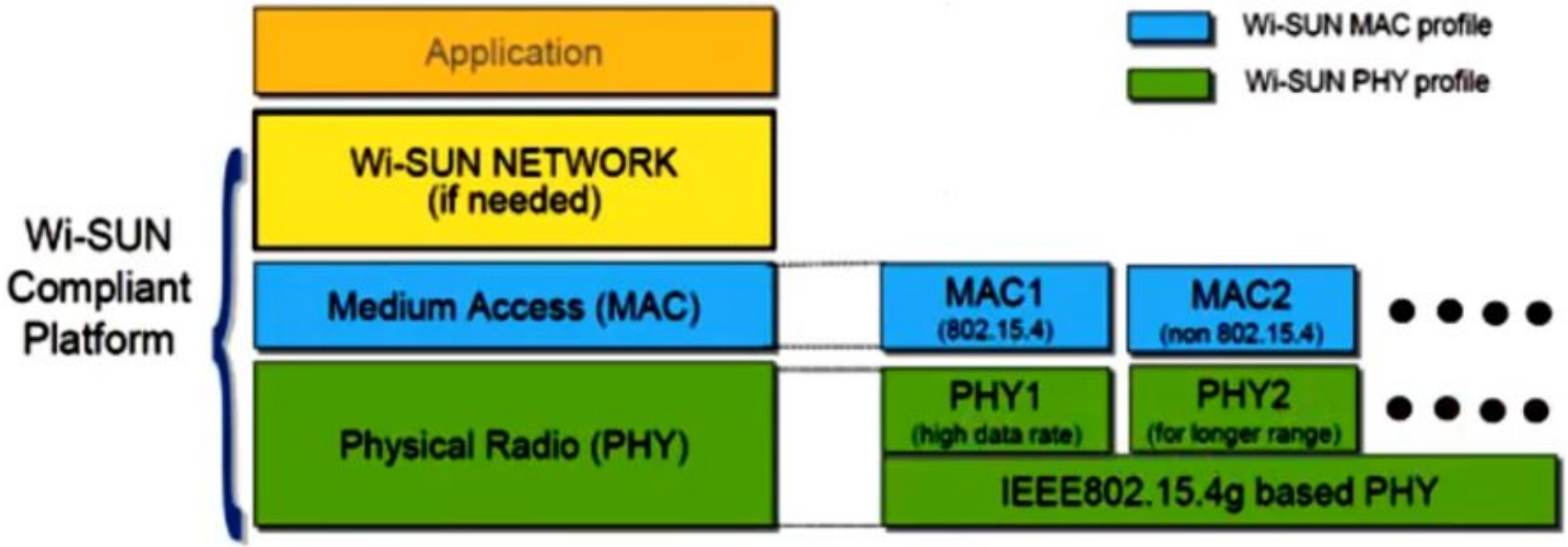
5. MAC Performance Metric: It is a mechanism to provide appropriate feedback on the channel quality to the networking and upper layers, so that appropriate decision can be taken.

6**. Fast Association (FastA**) The 802.15.4 association procedure introduces a significant delay in order to save energy. For time-critical application latency has priority over energy efficiency.

IEEE 802.15.4g

- IPv6 based Wireless Smart Utility Network (Wi-SUN) based on IEEE 802.15.4g
- Also known as Smart Utility Networks (SUN), was approved by IEEE in March, 2012
- Initially Japan focused, now expanding globally (US, South East Asia, India, Europe)
- Target smart utility use cases: Gas metering; demand/response; distribution automation

IEEE 802.15.4g

- PHY layer based on 4g but specification will be categorized based on use cases
- Frequency: 868 MHz (EU), 915 MHz (USA), 2.4 GHz ISM bands (worldwide)
- MAC may be based on or not based on 802.15.4. Application dependent

IEEE 802.15.4g